NØthing is Logical

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Slides: https://www.marialoni.org/resources/1BNU26.pdf











Beijing Normal University

Lecture 1, 12 October 2026

NØthing is logical (Nihil)

Outline of the lectures

1. **General introduction: Neglect-zero & no-split** [Mon, 12 Oct]
2. Modelling neglect-zero in tems semantics [Fri, 16 Oct]
3. Experimenting with zero-models [Mon, 19 Oct]

Additional lecture

- Learning how to split [Tsinghua, 15 Oct]

Literature

- Aloni, Maria (2022). "Logic and conversation: The case of free choice". In: *Semantics and Pragmatics* 15.5, pp. 1–60.
- Degano, Marco et al. (2025). "The ups and downs of ignorance". In: *Natural Language Semantics* 33, pp. 1–41.
- Klochowicz, Tomasz, Giorgio Sbardolini, and Maria Aloni (2026). "Cognitive bias approach to the acquisition of disjunction". In: *Proceedings of Sinn und Bedeutung. Vol. 30*, pp. 554–571.

NØthing is logical (Nihil)

- Goal of the project: a formal account of a class of natural language inferences which deviate from classical logic
- Common assumption: these deviations are not logical mistakes, but consequence of pragmatic enrichments (Grice)
- Initial strategy: develop *logics of conversation* which model next to literal meanings also pragmatic factors and the additional inferences which arise from their interaction
- Novel hypothesis: **neglect-zero** tendency (a cognitive bias rather than a conversational principle) as crucial factor
- Main conclusion: deviations from classical logic consequence of enrichments albeit not (always) of the canonical Gricean kind
  - Dominant view: enrichments always product of **rational interactions** between cooperative language users
  - Nihil hypothesis: enrichments sometimes result from our preference to minimize cognitive effort (**bias-driven** vs product of rational choices)

Nihil team


Anttila


Degano


Klochowicz


Knudstorp


Ramotowska


Yan


Zhou

...

Non-classical inferences

Free choice (FC)

(1) FC: $\Diamond(\alpha \vee \beta) \rightsquigarrow \Diamond\alpha \wedge \Diamond\beta$ [von Wright 1968]

(2) Deontic FC inference [Kamp 1973]

a. You may go to the beach *or* to the cinema.
b. $\rightsquigarrow$ You may go to the beach *and* you may go to the cinema.

(3) Epistemic FC inference [Zimmermann 2000]

a. Mr. X might be in Victoria *or* in Brixton.
b. $\rightsquigarrow$ Mr. X might be in Victoria *and* he might be in Brixton.

Ignorance

(4) The prize is either in the garden *or* in the attic $\rightsquigarrow$ The speaker doesn't know which [Grice 1989, p.45]

(5) ? I have two *or* three children.

- In the standard approach, ignorance is a conversational implicature
- Less consensus on free choice inferences analysed as conversational implicatures; grammatical (scalar) implicatures; semantic entailments; ...

The challenge of FC: adding FC to classical modal logic (ML) implies the equivalence of any two possibility claims

$$\Diamond a \Rightarrow_{\mathrm{ML}} \Diamond(a \vee b) \Rightarrow_{\mathrm{FC}} \Diamond b$$

Ignorance

The standard Gricean account

- Ignorance inferences are conversational implicatures
- Derived through implicit reasoning over what the speaker said and what could have said instead
- Example from Grice:

  (6) The prize is either in the garden *or* in the attic $\rightsquigarrow$ The speaker doesn't know which

- Sketch of Gricean derivation:

  The fact that a speaker uttered a weaker statement

  $$A \vee B$$

  when she could have produced stronger alternatives

  $$A; B$$

  means that it's not true that this speaker believes that the stronger statements hold

  $$\neg \Box_e A; \neg \Box_e B$$

Novel hypothesis: neglect-zero

- FC and ignorance inferences are [$\neq$ semantic entailments]
  - Not the result of Gricean reasoning [$\neq$ conversational implicatures]
  - Not the effect of applications of covert grammatical operators [$\neq$ grammatical (scalar) implicatures]
- They are rather a consequence of something else speakers do in conversation, namely,

  NEGLECT-ZERO
  when interpreting a sentence speakers construct models depicting reality (some verifying the sentence, some falsifying it) $\mapsto$ common assumption
  and in this process tend to neglect models that verify the sentence by virtue of an empty configuration (*zero-models*) $\mapsto$ novel hypothesis

勿空

- Tendency to neglect zero-models follows from the cognitive difficulty of:
  1. conceiving emptiness, the absence of things rather than their presence
  2. evaluating truths with respect to empty witness sets

[Nieder 2016; O. Bott *et al* 2019]

Novel hypothesis: neglect-zero

Illustration

(7) Less than three squares are black.

a. Verifier: [■, □, ■]
b. Falsifier: [■, ■, ■]
c. Zero-models: [□, □, □]; [■, ■, ■]; . . .

Zero-models in (7-c) verify the sentence by virtue of an empty set of black squares

- Cognitive difficulty of zero-models confirmed by experimental findings and connected to / can be argued to explain:
  - the special status of 0 among the natural numbers [Nieder 2016]
  - why downward-monotonic quantifiers are more costly to process than upward-monotonic ones (*less* vs *more*) [O. Bott *et al* 2019]
- NZ hypothesis: neglect-zero also at the origin of many common departures from classical reasoning
  1. FC and ignorance [MA 2022]
  2. Existential Import: every A is B ⇒ some A is B
  3. Principles from connexive logic: [Ziegler, Knudstorp & MA 2026]
     - Aristotle's Thesis: if not A then A ⇒ ⊥
     - Boethius' Thesis: if A then B & if A then not B ⇒ ⊥
  4. ESQ inferences & modified numerals [MA & vOrtmondt 2023]; Non-monotonic attitudes [Yan 2024]; SDA in counterfactuals [Klochowicz 2026], . . .

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(8) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(9) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(10) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(11) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. Zero-models: ?

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(12) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(13) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(14) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(15) M ate an apple or a banana.

a. Verifier: ?

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (15-c): verify the sentence by virtue of an empty witness for one of the disjuncts

Novel hypothesis: neglect-zero effects on disjunction

Illustrations

(16) Maria ate an apple.

a. Verifier: [🍎]

b. Falsifiers: [🍌]; [🍐]; [ ]

c. Zero-models: none

(17) Maria ate a banana.

a. Verifier: [🍌]

b. Falsifiers: [🍎]; [🍐]; [ ]

c. Zero-models: none

(18) M ate an apple and a banana.

a. Verifier: [🍎 🍌]

b. Falsifiers: [🍐]; [ ]

c. Zero-models: none

(19) M ate an apple or a banana.

a. Verifier: [🍎 | 🍌] ⇐ 'split'

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- Two **zero-models** in (19-c): sentence verified by virtue of an empty witness for one of the disjuncts
- **Split state** in (19-a): multiple (possibly conflicting) alternatives processed in a parallel fashion
- Neglect-zero hypothesis: Zero-models, where only one of the disjuncts is depicted, are cognitively taxing and therefore disregarded
- The exclusion of zero-models from the set of verifiers for a disjunction leads directly to an account of ignorance inferences and FC

A new conjecture: no-split

(20) Maria ate an apple or a banana.

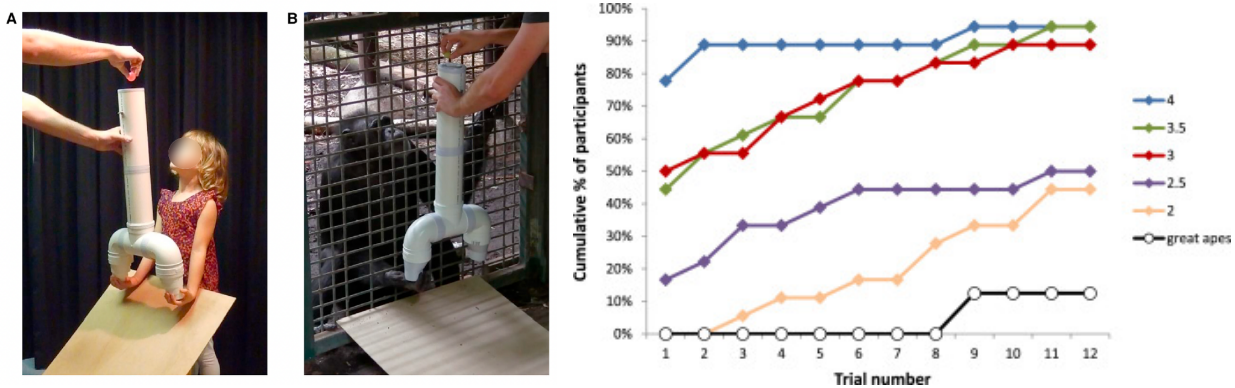
a. Verifier: [🍎 | 🍌] [⇐ **split state**]

b. Falsifiers: [🍐]; [ ]

c. **Zero-models**: [🍎]; [🍌]

- **Split states**: multiple (possibly conflicting) alternatives processed in a parallel fashion ↦ also a cognitively taxing operation

No-split conjecture [Klochowicz, Sbardolini & MA, SuB 2025]
the ability to split states (entertain multiple possibilities) is developed late



Preschoolers have trouble conceiving possible different futures [Redshaw & Suddendorf 2016]

- Proposal: Combination of neglect-zero + no-split can explain conjunctive interpretations of disjunction observed in pre-school children

No-split: conjunctive children

- Children (3-5/6y) sometimes (but systematically) interpret disjunctions conjunctively [Evans 93; Singh *et al* 16 (cf. Skordos *et al* 20); Cochard25; Bleotu *et al* 25]

(21) M ate an apple or a banana = M ate an apple and a banana
$\alpha \vee \beta \equiv \alpha \wedge \beta$

(22) M can eat an apple or a banana = M can eat an apple and a banana
$\Diamond(\alpha \vee \beta) \equiv \Diamond(\alpha \wedge \beta) \not\equiv \Diamond\alpha \wedge \Diamond\beta$

(23) M didn't eat an apple or a banana = M neither ate an apple nor a banana
$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$

- Proposal: children have conjunctive readings as they (similarly to adults) neglect zero and, unlike adults, lack the ability to split

1. Deriving ignorance:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{SPLIT}}$ [🍎 | 🍌]

   $\rightsquigarrow$ It might be an apple and it might be a banana (adults)

2. Deriving conjunctive reading:

   Apple OR banana $\Rightarrow_{\text{NZ}}$ 🍎 + 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [🍎 🍌]

   $\rightsquigarrow$ Both an apple and a banana (children)

3. In case of incompatible alternatives: [Leahy & Carey 2020, Bleotu *et al*]

   Left OR right $\Rightarrow_{\text{NZ}}$ ↙ + ↘ $\Rightarrow_{\text{NO-SPLIT}}$ **contradiction** ($\bot$)

   $\rightsquigarrow$ Sequential interpretation (children)

No-split: homogeneity

- Ability to split a state not only required to interpret disjunction but also conjunction under negation
- Neglect-zero + no-split can also derive homogeneus interpretations

[Sbardolini, 2023]

(24) M did *not* eat an apple *and* a banana $\rightsquigarrow$ M *neither* ate an apple *nor* a banana $\neg(\alpha \wedge \beta) \rightsquigarrow \neg\alpha \wedge \neg\beta$

- New conjecture: also adults tend to avoid split-states in cognitively taxing environments, notably under negation

1. Deriving homogeneity:

   NOT [apple AND banana] $\Rightarrow_{\text{NZ}}$ NO 🍎 + NO 🍌 $\Rightarrow_{\text{NO-SPLIT}}$ [NO 🍎 NO 🍌]

   $\rightsquigarrow$ Neither an apple nor a banana (children & no-splitting adults)

2. Deriving ignorance:

   NOT [apple AND banana] $\Rightarrow_{\text{NZ}}$ NO 🍎 + NO 🍌 $\Rightarrow_{\text{SPLIT}}$ [NO 🍎 | NO 🍌]

   $\rightsquigarrow$ Possibly not an apple and possibly not a banana (splitting adults)

- Testable predictions arising from these conjectures:
  - Increase of homogeneus (no-split) interpretations under cognitive load in adults (testable via dual-task studies)
  - Correlation between failure in Redshaw & Suddendorf's task and conjunctive behaviour in children

Cognitive bias approach (NIHIL$^+$)

Common assumption: Reasoning and understanding of natural language involve the creation of mental models [Johnson-Laird 1983, Mascarenhas & Koralus 2013, a.o.]

- **Understanding** a sentence S means being able to mentally construct a model picturing the world which verifies S, and possibly also a model which falsifies it
- **Reasoning** depends on two main processes: first construct verifying models for the premises and then check the validity of the conclusion on these models

Novel hypothesis: biases can constrain the construction of these models and therefore impact both reasoning and interpretation:

- NEGLECT-ZERO prevents the constructions of zero-models;
- NO-SPLIT expresses a dispreference for split-states;
- ...

Cognitive bias approach (NIHIL$^+$)

Comparison with competing accounts

	Ignorance	**FC & DIST**	**ES-Quant**	**Scalar impl.**	**Conj *or* & homog.**
Neo-Gricean	reasoning	reasoning	reasoning	reasoning	—
Grammatical	debated	grammar	grammar	grammar	grammar
NIHIL$^+$	neglect-zero	neglect-zero	neglect-zero	—	negl-z + no-split

- Standard derivation of **scalar implicatures** [Grice 75, Horn 72, ect]

(25) Maria ate an apple or a banana $\rightsquigarrow$ not both

a. Literal meaning: $\alpha \vee \beta$ [inclusive]
b. ALT: $\alpha \wedge \beta$
c. Negation of ALT: $\neg(\alpha \wedge \beta)$ [exclusive]

- NIHIL$^+$: **conversational inferences** sometimes bias-driven (*fast, effortless*) rather than product of alternative-based reasoning (*slow, costly*)

NEXT

- Logical modelling of biases in team semantics
- Experimental findings
  1. Degano *et al* (Nat Lang Sem, 2025): ignorance ⇐
  2. NIHIL *et al* (SuB24-25, CogSci25, XPRAG25): scalar, DIST & ES-Quant
  3. Bleotu *et al* (TbiLLC 2025, SuB25): on conjunctive *or*

Modelling biases in team semantics

General methodology

Natural language sentences translated into classical logic formulas interpreted in a team semantics which models both classical and bias-driven interpretations

NAT LANGUAGE $\mapsto$ LOGICAL LANG

α	**(classical)**	$\Rightarrow$
		$\Rightarrow$ Gric
		$\Rightarrow$ ean
		$\Rightarrow$ Prag
$[\alpha]^+$	(negl-zero)	$\Rightarrow$ mat
$[\alpha]^{++}$	(no-split)	$\Rightarrow$ ics
...		$\Rightarrow$

Back to FC challenge

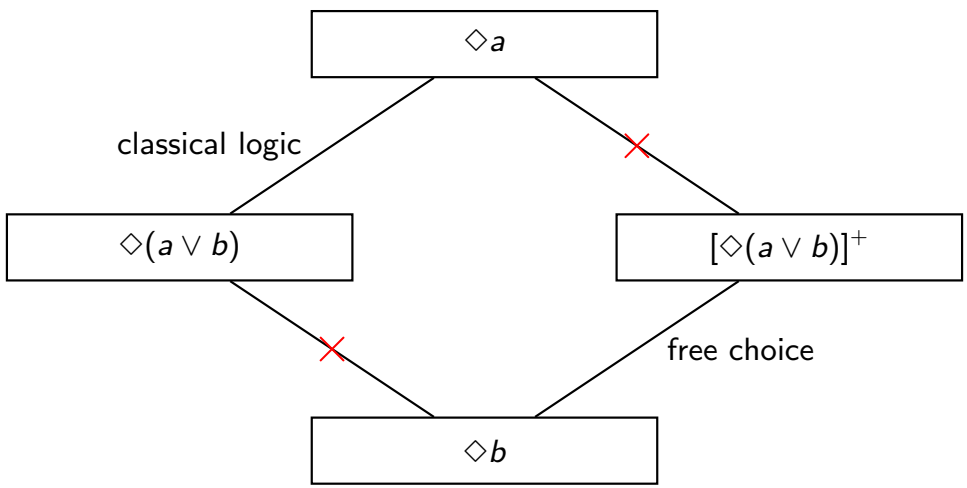


Figure: FC derived only for neglect-zero (NZ) enriched formulas

Modelling biases in team semantics

Team semantics

- Formulas interpreted wrt a set of points of evaluation (a team) rather than single ones [Hodges 1997; Väänänen 2007]
  - Classical modal logic: $[M = \langle W, R, V \rangle]$

$$M, w \models \phi, \text{ where } w \in W$$

  - Team-based modal logic:

$$M, t \models \phi, \text{ where } t \subseteq W$$

- Two crucial features
  - The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ zero-models
  - Multi-membered teams can model parallel processing of alternatives $\mapsto$ split states

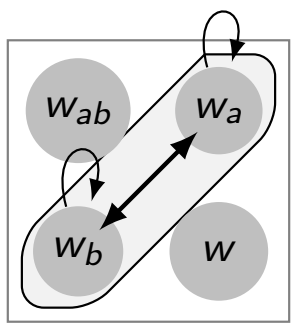


Figure: A multi-membered team modelling the parallel processing of possibilities a & b

Modelling biases in team semantics

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  - The empty set is among the possible teams ($\emptyset \subseteq W$) $\mapsto$ zero-models
  - Multi-membered teams can model parallel processing of alternatives $\mapsto$ split states

Neglect-zero & no-split

- Model-theoretic method: biases modelled by disallowing empty (neglect-zero) and multi-membered teams (no-split)
- Syntactic method: biases modelled via new logical atoms/operators
  - Neglect-zero: via **non-emptiness atom** NE which disallows empty teams as possible verifiers [Yang & Väänänen 2017]

$$M, t \models \text{NE} \text{ iff } t \neq \emptyset$$

  - No-split: via **flattening operator** F which induces pointwise evaluations and therefore mimics ban on parallel processing of alternatives

$$M, t \models \text{F}\phi \text{ iff for all } w \in t : M, \{w\} \models \phi$$

Modelling biases in team semantics

Classical and enriched interpretations

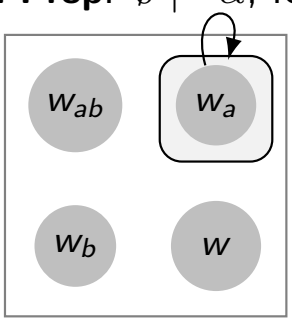
- $\alpha \Rightarrow$ empty team & split allowed $\mapsto$ **classical (literal) meaning**
- $[\alpha]^+ \Rightarrow$ empty team not allowed $\mapsto$ **NZ-enriched**
- $[\alpha]^{++} \Rightarrow$ neither empty team nor split allowed $\mapsto$ **NZ+NS-enriched**

Tensor disjunction in team semantics

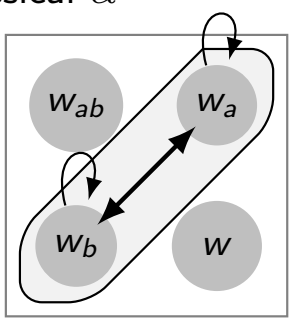
- Team t supports a **disjunction** iff t is the union of two subteams, each supporting one of the disjuncts
  - $a \vee b$: empty team allowed $\mapsto$ subteams can be empty (**classical**)
  - $[a \vee b]^+$: empty team not allowed $\mapsto$ subteams cannot be empty (**NZ-enr**)
  - $[a \vee b]^{++}$: neither empty team nor split allowed $\mapsto$ no empty or distinct subteams allowed (**NZ+NS-enriched**)

Empty Team Prop: $\emptyset \models \alpha$, for all classical α

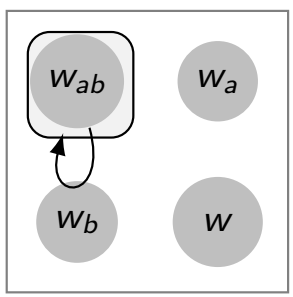
[**PROP**$= \{a, b\}; W = \{w_{ab}, w_a, w_b, w\}$]



$\models a \vee b$
$\not\models [a \vee b]^+$
$\not\models [a \vee b]^{++}$
$\mapsto$ **Zero-model**



$\models a \vee b$
$\models [a \vee b]^+$
$\not\models [a \vee b]^{++}$
$\mapsto$ **No-zero**, **split**



$\models a \vee b$
$\models [a \vee b]^+$
$\models [a \vee b]^{++}$
$\mapsto$ **No-zero**, **no-split**

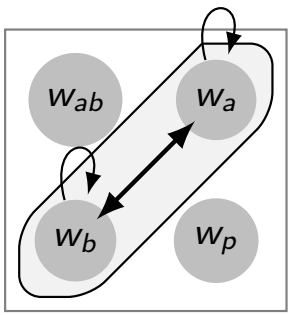
Modelling biases in team semantics: ignorance & conjunctive 'or'

(26) Maria ate an apple or a banana. $[a \vee b]$

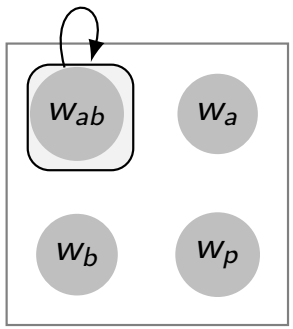
a. **No-zero**, **split** verifier: [🍎 | 🍌] $\rightsquigarrow$ it might be apple, it might be banana

b. **No-zero**, **no-split** verifier: [🍎 🍌] $\rightsquigarrow$ both apple and banana

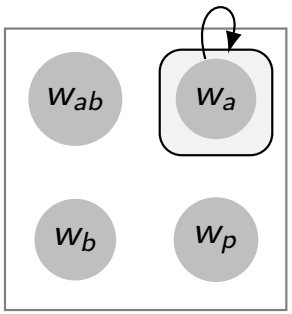
c. **Zero-models**: [🍎] ; [🍌] d. Falsifier: [🍐] $\rightsquigarrow$ neither apple nor banana



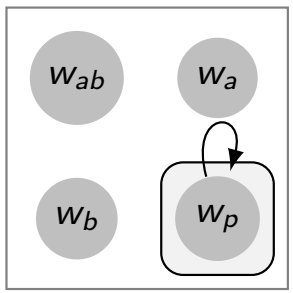
No-zero, **split** verifier



No-zero, **no-split** verifier



Zero-model



Falsifier

Predictions

We derive ignorance & conjunctive 'or' interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(27) M ate an apple or a banana $\rightsquigarrow_{nz}$ It might be an apple and it might be a banana

$[\alpha \vee \beta]^+ \models \Diamond_e \alpha \wedge \Diamond_e \beta$ (where R is state-based)

(28) M ate an apple or a banana $\rightsquigarrow_{ns}$ M ate both apple and banana

$[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$

(29) M didn't eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M ate neither apple nor banana

$[\neg(\alpha \vee \beta)]^{(++)} \models \neg\alpha \wedge \neg\beta$

and more predictions including FC, homogeneity, ...

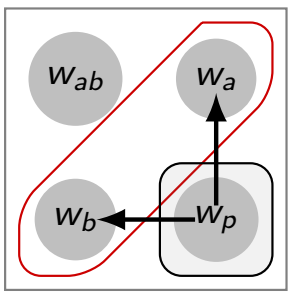
Modelling biases in team semantics: free choice

(30) Maria may eat an apple or a banana. $[\Diamond(a \vee b)]$

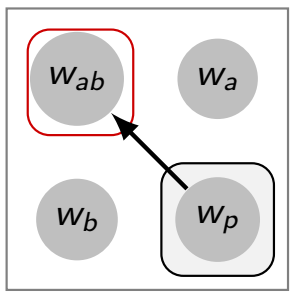
a. **No-zero**, **split** verifier: [🍎 | 🍌] $\rightsquigarrow$ may eat apple, may eat banana

b. **No-zero**, **no-split** verifier: [🍎 🍌] $\rightsquigarrow$ may eat both apple and banana

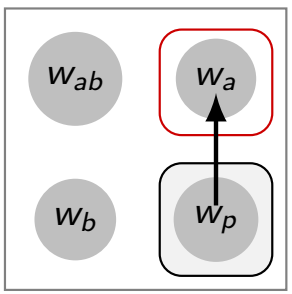
c. **Zero-models**: [🍎] ; [🍌] d. Falsifier: [🍐] $\rightsquigarrow$ may eat neither



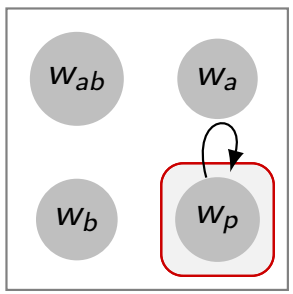
No-zero, **split** verifier



No-zero, **no-split** verifier



Zero-model



Falsifier

Predictions

We derive free choice & conjunctive permission interpretations for nz- and ns-enriched formulas resp. with no undesirable effects under negation

(31) M may eat an apple or a banana $\rightsquigarrow_{nz}$ It may be an apple and it may be a banana

$[\Diamond(\alpha \vee \beta)]^{+} \models \Diamond\alpha \wedge \Diamond\beta$

(32) M may eat an apple or a banana $\rightsquigarrow_{ns}$ M may eat both apple and banana

$[\Diamond(\alpha \vee \beta)]^{++} \models \Diamond(\alpha \wedge \beta)$

(33) M may not eat an apple or a banana $\rightsquigarrow_{(nz/ns)}$ M may eat neither apple nor banana

$[\neg\Diamond(\alpha \vee \beta)]^{(++)} \models \neg\Diamond\alpha \wedge \neg\Diamond\beta$

$\Rightarrow$ Freedom of choice requires ability to conceive possible alternative futures

A closer look at these predictions

Many enrichments triggered by disjunction

(34) Maria ate an apple or a banana $(\alpha \vee \beta) \rightsquigarrow$

a. Ignorance: speaker doesn't know which
b. Conjunctive interpretation: both $(\alpha \wedge \beta)$
c. Exclusive interpretation: not both $\neg(\alpha \wedge \beta)$

Can we capture all these enrichments as nz- or ns-effects?

Two components of full ignorance: possibility vs uncertainty

(35) Maria ate an apple or a banana $\rightsquigarrow$ speaker doesn't know which

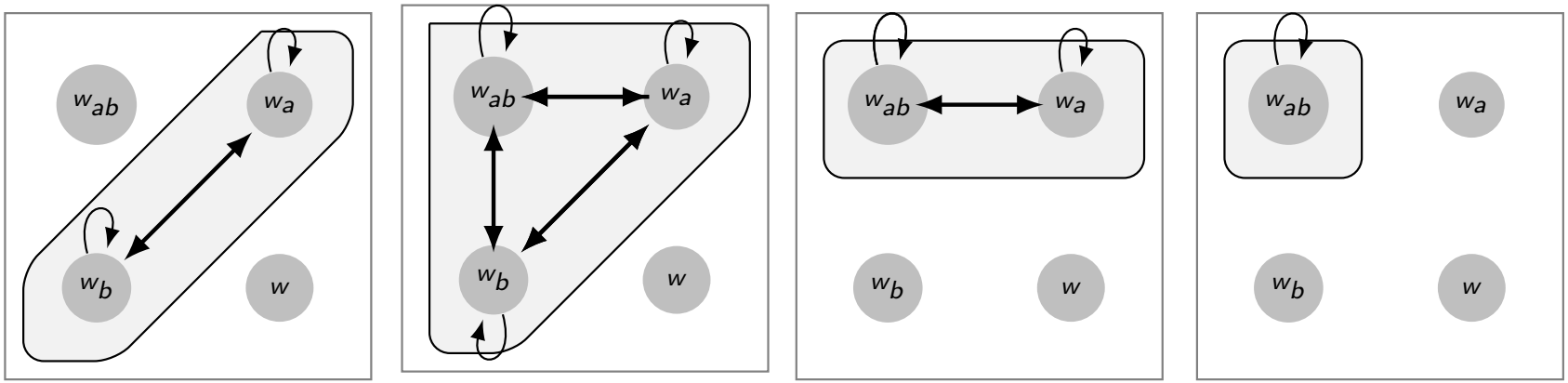
[Degano *et al* 2025][1]

a. Possibility: It is possible that M ate an apple and it is possible that M ate a banana $\Diamond_e\alpha \wedge \Diamond_e\beta$
b. Uncertainty: It is uncertain that M ate an apple and it is uncertain that M ate a banana $\neg\Box_e\alpha \wedge \neg\Box_e\beta$

[1] Degano, Marty, Ramotowska, MA, Sudo, Romoli, Breheny. The ups and downs of ignorance. *Natural Language Semantics* (2025)

Neglect-zero effects on disjunction: predictions of $[\ ]^{+\cdots}$-enrichment

Many no-zero verifiers for nz-enriched disjunction



(a) no-zero & excl $\models \neg(a \wedge b)$ (b) no-z & non-excl $\not\models \neg(a \wedge b)$ (c) no-z & no-uncertain $\not\models \neg\Box_e a$ (d) no-zero & no-split $\models (a \wedge b)$

Figure: Models for enriched $[a \vee b]^+$.

1. Neglect-zero enrichment derives **possibility**: $[\alpha \vee \beta]^+ \models \Diamond_e\alpha \wedge \Diamond_e\beta$
2. Neglect-zero + no-split derives conjunctive *or*: $[\alpha \vee \beta]^{++} \models \alpha \wedge \beta$
3. Neglect-zero enrichment does not derive exclusive readings: $[\alpha \vee \beta]^+ \not\models \neg(\alpha \wedge \beta)$
4. Neglect-zero enrichment does not derive uncertain inferences: $[\alpha \vee \beta]^+ \not\models \neg\Box_e\alpha$
   $\mapsto$ in contrast to standard neo-Gricean approach to ignorance

Two derivations of full ignorance

1. **Standard neo-Gricean derivation** [Sauerland 2004]

   (i) Uncertainty derived through quantity reasoning

   (36) $\alpha \vee \beta$ ASSERTION

   (37) $\neg\Box_e\alpha \wedge \neg\Box_e\beta$ UNCERTAINTY (from QUANTITY)

   (ii) Possibility derived from uncertainty and quality about assertion

   (38) $\Box_e(\alpha \vee \beta)$ QUALITY ABOUT ASSERTION

   (39) $\Rightarrow \Diamond_e\alpha \wedge \Diamond_e\beta$ POSSIBILITY

2. **Neglect-zero derivation**

   (i) Possibility derived as neglect-zero effect

   (40) $\alpha \vee \beta$ ASSERTION

   (41) $\Diamond_e\alpha \wedge \Diamond_e\beta$ POSSIBILITY (from NEGLECT-ZERO)

   (ii) Uncertainty derived from possibility and scalar reasoning

   (42) $\neg(\alpha \wedge \beta)$ SCALAR IMPLICATURE

   (43) $\Rightarrow \neg\Box_e\alpha \wedge \neg\Box_e\beta$ UNCERTAINTY

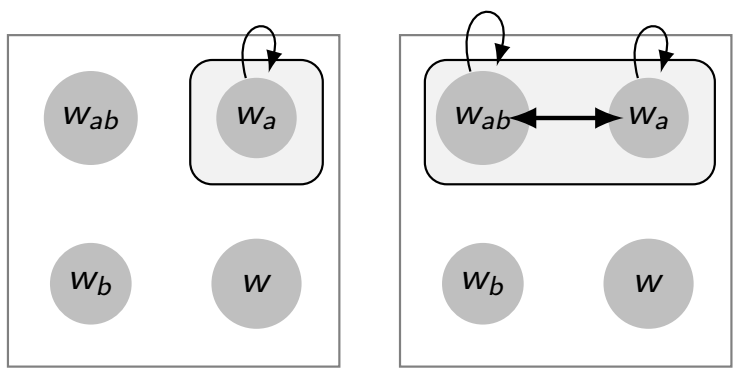
The ups and downs of ignorance [Degano et al 2025]

Contrasting predictions of competing accounts of ignorance

- Neo-Gricean: No possibility without uncertainty
- Neglect-zero: Possibility derived independently from uncertainty

Experimental findings

- Using adapted mystery box paradigm, compared conditions in which
  - both uncertainty and possibility are false [**zero-model**]
  - uncertainty false but possibility true [no-zero, no-uncertain model]
- Zero-models far more rejected (66%) than no-uncertain model (5%)
  $\Rightarrow$ Evidence that possibility can arise without uncertainty
- A challenge for the traditional neo-Gricean approach



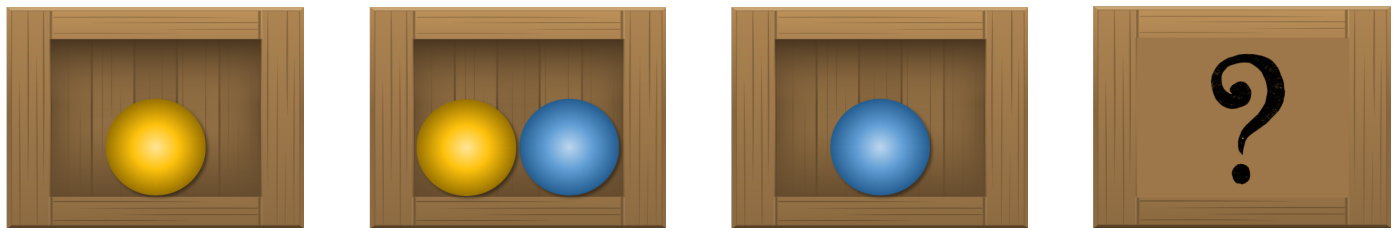
(a) **zero-model** (b) no-uncertain

Figure: Models for $(a \vee b)$

The mystery box paradigm[2]

The display

3 overt boxes containing colored balls and 1 mystery box.



The rule

The mystery box has the same contents as one of the overt boxes.

Examples

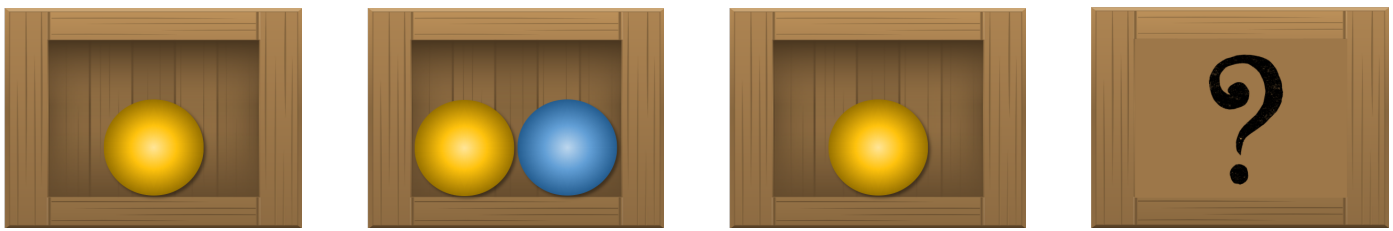
'The mystery box contains a yellow ball or a blue ball.' $Y \vee B$ **True**

'The mystery box contains a yellow ball or a green ball.' $Y \vee G$ **False**

[2] Adapted from Noveck 2001; see also Crnic et al. 2015.

Target-1 (no-uncertain)

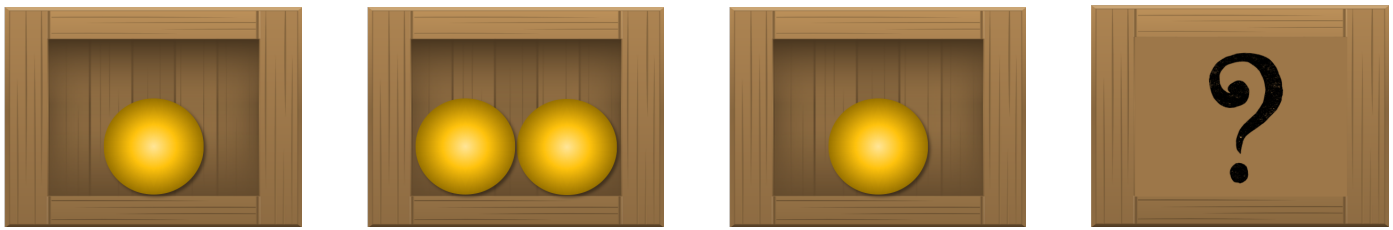
'The mystery box contains a yellow ball or a blue ball.' $Y \vee B$



UNCERTAINTY	$\neg\Box_e Y \wedge \neg\Box_e B$	**False**
POSSIBILITY	$\Diamond_e Y \wedge \Diamond_e B$	**True**

Target-2 (zero-model)

'The mystery box contains a yellow ball or a blue ball.' $Y \vee B$



UNCERTAINTY	$\neg\Box_e Y \wedge \neg\Box_e B$	**False**
POSSIBILITY	$\Diamond_e Y \wedge \Diamond_e B$	**False**

Hypothesis

If POSSIBILITY cannot arise without UNCERTAINTY, there should be no difference in responses between TARGET-1 and TARGET-2.

Results

Target-2 far more rejected (66%) than Target-1 (5%)

Conclusions

- Evidence that possibility can arise without uncertainty
- A challenge for the traditional neo-gricean approach

Conclusions

- FC, ignorance inference: a mismatch between logic and language
- Grice's insight:
  - stronger meanings can be derived paying more "attention to the nature and importance to the conditions governing conversation"
- Nihil hypothesis: some strengthenings due to cognitive tendencies rather than rational choices
  - FC, possibility and related inferences as neglect-zero effects

    Literal meanings (classical fragment) + cognitive factor (NZ) $\Rightarrow$ FC, possibility, connexive principles, etc
  - Conjunctive *or*, homogeneity as no-zero + no-split effect

    Literal meanings (classical fragment) + cognitive factors (NZ, NS) $\Rightarrow$ conjunctive *or*, homogeneity
- Implementation in team-based modal logics
- Ups and downs of ignorance: a challenge for the neo-Gricean approach

THANK YOU![3]

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